

SMART INDIA HACKATHON 2026 • PROBLEM STATEMENT SIH26104

SONIX

National Submission Pack

Everything the project is, everything it is not yet, and exactly what goes on each of the six slides.
Real-time detection of AI-cloned voices in live calls.
AICTE Cyber Security Cell.

Team **SONIX** • Yugal (ML pipeline & live engine) • Navya (augmentation, IndicVoices, ASVspoof 5) • Akshat (IndicSynth, In-the-Wild) • Anubhav (data extraction, mic capture, clip set) • Suryansh (demo UI, extension) • Yukti (metrics, presentation)

Branch `yugal/ml-pipeline` • Compiled 13 September 2026
Supersedes `SONIX_MASTER.md` for anything to do with the deck.

0. How to read this, and the two rules that govern everything

This document has six parts. **Part A** is the project A–Z. **Part B** is the six-slide build spec. **Part C** is the diagram library. **Part D** is what the research says actually wins at the national round. **Part E** is the Q&A bank. **Part F** is what must be run before submission.

RULE 1 — NEVER PUT A NUMBER ON A SLIDE THAT IS NOT IN THE LEDGER

§A6 is the ledger. A figure marked  is safe. Everything else is `[[PLACEHOLDER]]` until its owner fills it in. At the national round a jury of three visits you across three scoring rounds and one of their documented questions is “What is your accuracy, and how did you measure it?” A number you cannot decompose on demand costs you the credibility of every number beside it.

RULE 2 — THE METRIC IS EER, AND HIGHER SCORE MEANS THE MODEL THINKS IT IS FAKE

Our benchmark data is roughly 90 % spoof. A model that answers “fake” every single time scores 90 % accuracy and is worthless. Say **EER**, and say the direction of the score out loud the first time one appears on screen — judges misread the axis otherwise.

Correction to earlier guidance

Previous versions of our planning documents cited an evaluation rubric of *Innovation 25 % / Problem understanding 20 % / Technical feasibility 20 % / Impact 20 % / Presentation 15 %*, and said the deck could run to ten slides. **Both came from an SEO blog and both are wrong.** SIH publishes no percentage weights anywhere, and the official template caps the deck at six slides. The corrected facts are in Part D. Anything built on the old numbers needs re-checking.

PART A — The project, A to Z

A1. What SONIX is, in one paragraph

Voice cloning is now cheap and convincing from seconds of audio. Attackers impersonate CXOs, bank officials and family members to authorise transfers, extract OTPs and bypass voice verification. India is a large target: UPI fraud, family-emergency scams, official impersonation. **Telephony has no mechanism to tell a cloned voice from a real one while the call is happening.** SONIX scores a live call for the probability that the voice is synthetic and surfaces it as Green / Amber / Red. It never blocks a call — it flags for a human.

A2. What the problem statement demands

Three of these are quoted because they drive design decisions we have to defend:

*“expose **APIs and SDKs** for seamless integration with banking applications, enterprise communication systems, and telecom operator infrastructures”*

*“**REST/gRPC APIs and SDKs** for integration with core banking systems, contact center platforms, enterprise communication tools, and telecom networks”*

*“A **reusable security layer** for telecom operators and enterprises”*

“diverse Indian accents and dialects”

The first three are why SONIX is built as middleware rather than an app (\$A7). The fourth is why the entire Indic effort exists (\$A8, case 3).

A3. The pipeline

```
live audio (16 kHz mono)
→ 4.0 s windows at 0.5 s hop          87.5 % overlap
→ silence gate (VAD)                  near-silent windows dropped
→ wav2vec2 XLS-R 300M, FROZEN          → mean-pool → 1024-dim
→ MLP head, TRAINED                   1024 → 256 → ReLU → Dropout(0.3) → 1
                                       262,657 parameters
→ sigmoid                             → P(synthetic), 0–1
→ 5-window moving average
→ hysteresis: 3 of 5 must agree to switch band
→ GREEN / AMBER / RED
```

WHY THIS SHAPE — THE ARCHITECTURE ARGUMENT

Frozen front-end, tiny trainable head. 300 million frozen parameters listen; 262,657 trained parameters decide. When a new cloning tool appears: extract embeddings once, retrain the head in minutes, serve it from `/api/models`. No redeployment, no downtime, no second GPU — every head shares the one resident front-end.

Embeddings are the expensive reusable artifact. Extracting them is hours of GPU time; training a head on them is minutes. This is also why the team's work divides as it does — teammates produce embeddings, one machine trains heads.

Streaming, not batch. Overlapping windows, a silence gate, temporal smoothing and hysteresis. This is the machinery that makes a detector deployable rather than a notebook result, and almost no published detector has it.

A4. Thresholds and band logic

SETTING	VALUE	WHY
Amber	0.10	data-driven from the eval score distribution
Red	0.90	same
Smoothing	5-window moving average	one odd window cannot flip the verdict
Hysteresis	3 of 5 must agree to switch	prevents band flicker on the operator's screen

Both thresholds are operator-configurable. A bank sets them stricter for fund transfers than for general enquiries — say this out loud in the demo, it is a deployment-realism point and those score.

A5. Data

TRAINING CORPUS — WHAT THE CURRENT HEAD ACTUALLY SAW

EMBEDDING ROOT	BONAFIDE	SPOOF
<code>embeddings</code> — ASVspoof 2019 LA, clean	2,580	22,800
<code>embeddings_g711</code> — G.711 μ -law codec copies	2,580	22,800
<code>embeddings_rawboost</code> — RawBoost channel/impulsive	2,580	22,800

<code>embeddings_rirmusan_*</code> — room reverb + MUSAN noise	2,580	22,800
<code>embeddings_indicvoices_tr</code> — genuine Indian speech	37,032	0
<code>embeddings_mms_tts + _aug</code> — Indic TTS spoof	0	1,600
<code>embeddings_indicsynth + _aug</code> — Indic VC spoof	0	33,600
TOTAL	47,352	126,400

The Indic slice is 37,032 genuine to 35,200 synthetic — a 1.05 : 1 ratio. Before this work it was 37,032 to **zero**. That ratio was the bug.

THE INDIC SPOOF CORPUS WE BUILT — 35,200 EMBEDDINGS, 3 SYNTHESIS FAMILIES, 12 LANGUAGES

FAMILY	ROWS	LANGUAGES
MMS-TTS (text-to-speech)	1,600 — 800 clean + 800 channel-augmented	Hindi, Marathi, Bengali, Tamil, Telugu
IndicSynth (voice conversion)	33,600 — 16,800 + 16,800 augmented	12 Indian languages

Full language list: Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Odia, Punjabi, Sanskrit, Urdu. Channel augmentation applied to half of each family: G.711 μ -law round-trip, synthetic RIR convolution reverb, pink noise at 8–30 dB SNR.

HELD-OUT TEST SETS — NEVER TRAINED ON

- ASVspoof 2019 LA **eval** — 71,237 clips (7,355 bonafide / 63,882 spoof)
- ASVspoof 2019 LA **dev** — 24,844 (2,548 / 22,296)
- ASVspoof 2021 **DF** — 75 % CM-key coverage, and that caveat must appear beside any DF21 figure
- **In-the-Wild** — real internet deepfakes, tools nobody told us about
- **IndicVoices holdout** — 12 % of recordings, split by *recording id* not row id, so `<id>_chunk_N` siblings cannot leak across the split
- `sonix_real/` — our own labelled real-and-cloned recordings. See below.

SONIX_REAL/ — THE SET THAT DECIDES EVERYTHING

Real and cloned recordings of the same Indian public figures: 9 named genuine (Modi, Rahul Gandhi, SRK, Bhuvan Bam, Ashish Chanchlani, Manmohan Singh, Nizamuddin, Rajat Sood, TVK Vijay), 10 named clones of the same speakers, plus 10 numbered English real/fake pairs recorded by the team on phone microphones and cloned with free tools.

This is the only test that answers the question the whole Indic effort exists for: *can we tell a cloned Indian voice from the real one?* **A matched real/clone pair from the same speaker is also the single most persuasive thing in the demo** — nothing else lands the way it does.

A6. The numbers ledger

LOCKED — SAFE TO PRESENT

FACT	VALUE
Architecture	frozen wav2vec2 XLS-R 300M + 262,657-parameter MLP head
Windowing	4.0 s window, 0.5 s hop, 87.5 % overlap
Band logic	5-window moving average, hysteresis 3-of-5
Thresholds	Amber 0.10, Red 0.90, operator-configurable
API surface	7 REST endpoints + WebSocket, 2 independent clients
Padding bug — 1 second	0.8795 → 0.0329
Padding bug — 2 seconds	0.5061 → 0.0005
Padding bug — 3 seconds	0.2224 → 0.0096
Padding bug — full 66 s (control)	0.1171 → 0.1171, unchanged
SNR shortcut — genuine clips that passed	34.6, 37.7 dB → median 0.011, 0.002
SNR shortcut — genuine clips that failed	39.2–66.8 dB → median 0.779–1.000
SNR shortcut — synthetic clips	96–140 dB
Indic spoof corpus built	35,200 embeddings , 3 generator families, 12 languages
Training corpus total	47,352 bonafide / 126,400 spoof
Calibration temperature	$T \approx 1.05$
Calibration effect on EER	none — identical before and after, and it was never meant to change it

CONTESTED — MUST BE RE-ESTABLISHED BEFORE IT GOES NEAR A SLIDE

ASVspoof-2019 LA eval EER **1.4937 %** at threshold 0.095130 for the baseline `head.pt`, eval set 71,237. Independently reproduced by two implementations to four decimals on 1 September. **But `head.pt` was later overwritten during a merge**, so this figure is unverified until `verify_head.py` passes. Treat it as pending.

PENDING — PLACEHOLDERS ONLY, EACH WITH AN OWNER

TOKEN	OWNER	BLOCKED ON
<code>[[EER_INDOMAIN]]</code>	Yugal	<code>verify_head.py</code> on the current checkpoints
<code>[[EER_DF21]]</code>	Navya	final model — must carry “(75 % key coverage)”
<code>[[FA_REAL]]</code>	Yugal	final model + <code>sonix_real/</code>
<code>[[FA_INDIC]]</code>	Navya	final model. Per 4-second chunk, not per call
<code>[[INDIC_CLONE_DETECT]]</code>	Yugal	the <code>sonix_real/</code> pair test — unblocked, just not run
<code>head_v3</code> real EER	Yugal	dev EER 0.24 % is a sanity check, never a headline

head_full_indic_as5 anything	Navya	nothing has been verified on our machine
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🚫 BLOCKED — DESIGN THE SLIDE TO SURVIVE WITHOUT

TOKEN	WHY	PLAN B
[[EER_ITW]]	In-the-Wild not extracted	“🕒 measurement in progress” row in the benchmark matrix
Cost / scale figures	nobody assigned	30 minutes of work — see \$F. Winners cite this as a differentiator
Latency	never measured	do not quote a number, in any round

THE CLAIM THAT MUST NOT APPEAR

We measured **57 % of genuine Indian speech flagged as fake**, then trained a model that cut it to **0.02 %**. Both numbers are real. **But that model also passes cloned Indian voices as genuine** — every Indian sample it ever saw was real, so it learned “*Indian audio = real*” rather than learning to detect anything.

Say: “We discovered our detector fails on Indian speech, measured it, diagnosed the cause, and built a 35,200-clip Indian synthetic-speech corpus across three generator families to fix it.”

Never say: “We fixed it — 57 % to 0.02 %.” If the sonix_real/ test verifies before submission this moves to 🔒. Until then it does not.

✗ NEVER SAY, IN ANY ROUND

“99 % accurate” · “detects any AI voice” · any latency figure · “we have a production API” (say “**a working service interface**”) · any number not marked 🔒 above.

A7. Deployment — an API and SDK, not an application

SONIX is middleware. Banks, contact centres and telcos call it from inside systems they already run. The dashboard is a *client*, not the product.

LAYER	WHAT IT IS	HOW TO PRESENT IT
Scoring engine	frozen front-end + trained heads, batched across concurrent calls	the core IP
Integration API	7 REST endpoints + WebSocket, consent-gated	the deliverable
Operator console	Streamlit dashboard, live risk timeline	a reference client
Chrome extension	browser-call capture	a second client — proves platform-agnosticism

METHOD	PATH	PURPOSE
GET	/api/status	health, mode, active calls, engine stats
GET	/api/models	which heads are loaded, and their descriptions
POST	/api/score-file	submit audio; returns immediately, streams results
GET	/api/telemetry?call_id=	per-window scores, VAD decisions, ring-buffer stats
POST	/api/approve	approve the pairing / consent code

POST	/api/end-call	finalise and archive the call
WS	/ws	live score stream, and browser mic audio in

THE THREE LINES THAT PROVE WE READ THE PROBLEM STATEMENT

```
from sonix_sdk import SonixClient
sonix = SonixClient("http://sonix.internal:8000")

result = sonix.score_file("call_88213.wav")
if result.band == "RED":
    hold_transaction(reason=f"voice-clone risk {result.mean:.0%}")
```

This goes on slide 3. A judge who reads it believes the integration story instantly — and it is the single cheapest credibility win in the deck.

THE STRONGEST HONEST CLAIM WE HAVE

Two independent clients already consume the same endpoints — a Python dashboard and a JavaScript Chrome extension, sharing no code. That is the real test of whether something is an interface or just an app's backend, and most hackathon “APIs” fail it.

MATURITY — STATE IT YOURSELF BEFORE A JUDGE DOES

We have a working service interface: 7 REST endpoints, WebSocket streaming, proper status codes, two independent clients.

What it is not yet: authenticated, versioned, or a published SDK. Those are days of work, not architecture changes — the separation that makes them cheap already exists. Drawing that line yourself earns more credibility than claiming a finished API, and the research in Part D says so explicitly.

A8. The story spine — we find our own failures and fix the cause

This is what makes a practitioner jury believe everything else on the slide. Anyone can show a good number. Almost nobody shows a measured before-and-after of a bug they found themselves. **Frame all four as: observation → hypothesis → measurement → root-cause fix → verification.**

CASE 1 — THE PADDING BUG

The model listens in 4-second chunks. Shorter clips were padded with *digital silence*, so a 1-second clip was three-quarters nothing — and the model called real people fake. We now repeat the audio instead of padding it.

CLIP LENGTH	BEFORE (ZERO-PAD)	AFTER (REPEAT-PAD)
1 second	0.8795	0.0329
2 seconds	0.5061	0.0005
3 seconds	0.2224	0.0096
Full 66 s (control)	0.1171	0.1171 — unchanged

The control row is the part a sharp judge will respect: no padding is involved at 66 seconds, so it is unchanged — which is how we know the fix did only what we claim it did.

CASE 2 — THE BACKGROUND SHORTCUT

Genuine recordings still flagged. Instead of tuning thresholds we investigated: the model had learned “**clean background = fake**”, not synthesis detection. ASVspoof’s silence is digitally dead at -75 dBFS, a floor no real microphone produces. Phone noise-suppression strips room tone, so genuine voices looked synthetic. **Fixed in the training data** — room reverb and real-world noise — not in the thresholds.

CASE 3 — THE INDIAN-SPEECH BLIND SPOT, AND THE SECOND SHORTCUT UNDERNEATH IT

The PS demands “diverse Indian accents and dialects”, so we tested it. The majority of genuine Indian speech was flagged as fake. We added IndicVoices to training and the false-alarm rate collapsed. Then we tested a matched pair — a real and a cloned recording of the same speaker — and found the fix had created a *second* shortcut:

MODEL	REAL INDIAN VOICE	CLONED INDIAN VOICE	WHAT IT ACTUALLY LEARNED
head_robust_v2	✗ called fake	✓ called fake	“Indian audio = fake”
head_full_ho	✓ called real	✗ called real	“Indian audio = real”

The cause, verified in the extraction script: **IndicVoices is bonafide-only**. `grep -c spoof scripts/extract_indicvoices.py` returns 0 — one label constant, `BONAFIDE_LABEL = 0`. The only association available to the model was “Indian phonetics → genuine”. The 0.02 % false-alarm figure is an honest measurement and says **nothing whatsoever** about detection.

The fix: build Indian synthetic speech so the language appears on both sides of the label — 35,200 spoof embeddings across three families, bringing the Indic slice from 37,032 : 0 to 37,032 : 35,200. **Verification is the open item**, and the `sonix_real/` pair test is the gate.

CASE 4 — THE ENGINE BUGS (HOLD THIS ONE FOR THE ORAL ROUNDS)

Two production-shaped defects found and fixed during integration. Do not put these on a slide; deploy them when a judge asks about engineering rigour or testing.

- A `NameError` in the head forward pass killed the scoring task on the first batch. The server kept accepting audio and scored nothing, *silently*, because `asyncio` swallowed the exception. Fixed, plus a done-callback that shouts, plus per-batch isolation so one failure cannot end scoring for the process.
- `_trim_pending()` subtracted an int from `None` — uploads deliberately have no backlog cap — killing the audio feed on the *first* window of every upload. A 62-window clip returned one score, filed under index `-1`, and the dashboard drew a confident GREEN band from it.

The line worth saying: “A *detector that fails loudly is safer than one that fails silently, and we now instrument for that.*” It also explains why every upload result collected before 3 September is void, which is a fact we should volunteer rather than have discovered.

A9. The models

Seven heads are registered and present on disk. All share the one frozen front-end.

KEY	TRAINED ON	WHAT IS ACTUALLY VERIFIED
baseline	ASVspooof 2019 LA clean	1.4937 % eval EER — needs re-verification after a merge overwrite
augmented	+ G.711 μ -law	dev 0.24 % — not comparable to baseline, different test set
robust	+ RawBoost	—
robust_v2	+ RIR / MUSAN	learned “Indian audio = fake”
full_ho	+ IndicVoices, holdout-safe	learned “Indian audio = real”; DF21 5.27 % vs baseline 9.48 %
full_indic_as5	+ IndicVoices + ASVspooof 5 (Navya)	nothing verified on our machine
v3 ← default	everything + 35,200 Indic spoofs	dev EER 0.24 % @ epoch 31 — sanity check only

Why v3 is the default: it is the only head with Indian languages on *both* sides of the label. That is a reasoning-based choice, **not** a verified head-to-head. `head_full_ho` 's 0.02 % Indian false-alarm rate looks better and measures nothing. The `sonix_real/` pair test decides the real winner.

Why dev EER is never a headline: dev shares attack types with training, so a near-zero figure there is expected, not evidence of quality. **Sanity rules for any EER:** single digits = working · ~40 % = label bug · ~0 % = data leak.

A10. Current state of the build — as of 13 September

COMPONENT	STATE
Live streaming engine (windows, VAD, smoothing, hysteresis)	WORKING
7 REST endpoints + WebSocket	WORKING
Operator dashboard, per-window timeline	WORKING
Chrome extension (second API client)	WORKING
Consent gate + immutable audit trail	WORKING
7 trained heads, hot-swappable at runtime	WORKING
35,200-clip Indic synthetic corpus	BUILT & VERIFIED
<code>sonix_real/</code> matched pair test	NOT RUN

<code>verify_head.py</code> on current checkpoints	NOT RUN
Cross-dataset EER (DF21, In-the-Wild) on the final model	NOT RUN
Ablations (what each generator family bought)	NOT RUN
Cost / scale / latency figures	NOT MEASURED

The pipeline has not moved since 3 September. Everything in the red rows is a \$F item, and the first two are cheap.

PART B — The six-slide build spec

THE CONSTRAINT, VERBATIM FROM THE OFFICIAL TEMPLATE

“Kindly keep the maximum slides limit up to **six (6)**. (Including the title slide)” · “Try to avoid paragraphs and post your idea in **points / diagrams / infographics / pictures**” · “Idea should be **unique and novel**” · “You can **only use provided template** without changing the idea details pointers” · “You need to **save the file in PDF**. No PPT, Word Doc or any other format will be supported.”

You cannot add a slide. There is no room for a cost slide, a business-model slide, a team slide or an IP slide. Everything has to be compressed into the six fixed sections, and anything that will not fit is held for the oral rounds. Delete the template's “Important Pointers” slide before export — the template says to, and leaving it in is a documented way teams look careless.

Slide 1 — Title

Template fields, fill exactly as registered on the portal: Problem Statement ID `SIH26104` · PS Title · Theme · PS Category (**Software**) · Team ID · Team Name.

TWO DOCUMENTED WAYS TEAMS LOSE MARKS HERE

Theme metadata that does not match the portal entry, and template placeholder text left on the slide. Both are trivially avoidable and both are noticed. Cross-check every field against the portal before export.

No visual. Do not add a logo, a tagline or a hero image — the instruction says use the provided template without changing the pointers.

Slide 2 — Proposed Solution

Template asks for: detailed explanation of the proposed solution · how it addresses the problem · **innovation and uniqueness of the solution.**

Primary visual: Figure 1 (attack-and-defence flow). **Secondary, bottom third:** Figure 8 (the differentiator table) — this is what answers “innovation and uniqueness” in a form a reviewer can scan in four seconds.

HEADLINE

Every deepfake tool analyses the recording afterwards. SONIX scores the call while it is happening.

THE SIX BULLETS — DO NOT EXCEED SIX, DO NOT READ THEM ALOUD

- **Live, not post-call** — a verdict every 0.5 s on a 4 s rolling window, not a file analysed after the money moved
- **Flags, never blocks** — a false positive that cuts off a real emergency is a worse failure than the fraud
- **A reusable security layer** — 7 REST endpoints + WebSocket, already consumed by two independent clients
- **Retrains in minutes** — 300M frozen parameters listen, 262,657 trained parameters decide
- **Built for Indian languages** — 12 languages, and we built the synthetic half of that data ourselves
- **Consent and audit in the state machine** — audio is dropped in code until the call is approved

WHERE THE NOVELTY MARKS ARE, AND HOW TO CLAIM THEM

“Novelty of the idea” is the first official criterion, and “we trained a deepfake detector” is what every other team will write. Our four genuine differentiators, in the order they are worth marks: **(1)** streaming architecture with silence gate, smoothing and hysteresis — nearly every published detector is batch; **(2)** frozen front-end / 262k trainable head, so a new attack costs minutes not days; **(3)** consent and audit enforced in the state machine rather than promised in prose; **(4)** we found and root-caused two shortcuts in our own model. The fourth is the rarest and belongs on slide 4.

Slide 3 — Technical Approach

Template asks for: technologies to be used (programming languages, frameworks, hardware) · methodology and process for implementation (**flow charts / images / working prototype**). The template explicitly invites a flow chart here — take it.

Primary visual: Figure 2 (the pipeline), across the full width. **Secondary:** Figure 3 (integration architecture), bottom-left.

Bottom-right: the three-line SDK snippet.

HEADLINE

300 million frozen parameters listen. 262,657 trained parameters decide.

STACK, IN SMALL TYPE — NAME THE LAYER, NOT A LOGO WALL

LAYER	WHAT WE USED
Front-end (frozen)	wav2vec2 XLS-R 300M, PyTorch, HuggingFace transformers
Trained head	PyTorch MLP, 262,657 params, <code>pos_weight</code> class balancing
Serving	Python, aiohttp, REST + WebSocket, off-loop inference via <code>asyncio.to_thread</code>
Clients	Streamlit operator console; Chrome extension (JavaScript)
Augmentation	G.711 μ -law, RawBoost, RIR convolution, MUSAN
Hardware	trained on 3 consumer GPUs (1×RTX 3050, 2×RTX 4060)

Name the hardware. “Trained on three consumer GPUs” is a feasibility signal — it says this does not need a datacentre, which is exactly what “practicability” and “sustainability” in the official criteria are asking about.

THE SINGLE HIGHEST-LEVERAGE ELEMENT IN THE WHOLE DECK

The three-line SDK snippet (§A7). The problem statement asks for APIs and SDKs three times; most teams will present an app. A judge who reads those three lines believes the integration story instantly, and it costs you four centimetres of slide.

Slide 4 — Feasibility and Viability

Template asks for: feasibility analysis · **potential challenges and risks** · strategies for overcoming them. Note that the template *requires* you to state what is hard — honesty is a scored field here, not a weakness you are admitting.

This is our strongest slide. Primary visual: Figure 4 (the three shortcuts). If space allows, Figure 6 (padding chart) as a small inset.

HEADLINE

We audit our own model. Three shortcuts found, two fixed in the data, one in verification.

LEFT COLUMN — FEASIBILITY, DEMONSTRATED NOT ASSERTED

- Working end-to-end today: live audio → score → band, with 7 heads hot-swappable at runtime
- Trained on consumer GPUs — no datacentre required
- Self-hosted container; the call audio never leaves the organisation's network
- A new attack costs **minutes** of retraining, not a release cycle

RIGHT COLUMN — RISKS, AND WHAT WE DID ABOUT EACH

RISK	WHAT WE DID
Cross-dataset EER degrades — the known open problem in this field, typically 10–30 %	we measure it rather than avoid it, and the system flags rather than blocks precisely because of it
Model learns a shortcut instead of the task	found twice, root-caused twice, fixed in the training data both times — not by moving a threshold
Indian-language coverage	built a 35,200-clip synthetic corpus, 3 generator families, 12 languages
False positive on a genuine emergency call	never blocks; 5-window smoothing + 3-of-5 hysteresis; operator-set thresholds

THE BENCHMARK MATRIX — NEVER LEAVE A BLANK CELL

BENCHMARK	WHAT IT TESTS	STATUS
ASVspoof 2019 LA eval	in-domain accuracy	[[EER_INDOMAIN]]
ASVspoof 2021 DF	cross-dataset generalisation	[[EER_DF21]] (75 % key coverage)
In-the-Wild	real internet deepfakes, unseen tools	 in progress
IndicVoices holdout	language robustness	[[FA_INDIC]]
sonix_real/ matched pairs	real-world Indian clone detection	 in progress

Caption under the matrix: “Evaluation is ongoing. Cross-dataset EER for audio anti-spoofing is typically 10–30 % — degradation is the known open problem in this field, and we measure it rather than avoid it.”

Stating the 10–30 % expectation *first* inoculates you: a reviewer cannot then use it against you. Against five decks claiming 99 %, the one naming its open problems is the one a practitioner trusts.

Slide 5 — Impact and Benefits

Template asks for: impact on the target audience · benefits (social, economic, environmental).

Primary visual: Figure 11 (stakeholder map) or Figure 10 (deployment and privacy). **Pick one.** If the jury skews technical or government, use Figure 10 — “the audio never leaves their network” is the stronger argument for a cybersecurity PS.

HEADLINE

The score reaches a human while the call is still live — the only moment stopping this is still cheap.

ECONOMIC, WITH THE MECHANISM NAMED

- **Banks and contact centres** — hold a transfer before it clears. After it clears, recovery is a legal process; before, it is a click.
- **Telecom operators** — network-level screening offered as a subscriber service; one deployment protects millions of calls
- **Enterprises** — CXO payment fraud on internal calls, the fastest-growing vector

SOCIAL

- **Individuals** — family-emergency scams, which disproportionately target elderly and less digitally-literate users
- **Government helplines** — official impersonation, where the attack is the claim of authority
- **Language equity** — a detector that works on Indian languages, because one trained only on English benchmarks quietly fails most of the country

PRIVACY — GIVE IT REAL SPACE, IT IS A DIFFERENTIATOR ON A CYBERSECURITY PS

- Consent gate **enforced in the state machine** — `push_audio()` drops every frame until the call is approved. A code path, not a policy paragraph.
- Immutable per-call audit trail: what was scored, when, under which model
- **No audio retained** — only the score and the audit record leave the container. Aligns with the DPDP Act's data-minimisation principle by construction, not by promise.

Name one framework, not five. DPDP is the one that genuinely binds a system that touches call audio. A carpet of acronyms (CERT-In, TRAI, RBI, ISO) reads as padding, and “clarity” is an official scored criterion working against you.

Slide 6 — Research and References

Template asks for: details and links of the reference and research work.

Two columns. **Prefer peer-reviewed sources to blog links** — the 2024 winning deck on a security PS cited Elsevier and ScienceDirect, and it reads as a different level of seriousness.

DATASETS	PAPERS & METHODS
ASVspoof 2019 LA / 2021 DF / ASVspoof 5	Müller et al., <i>Does Audio Deepfake Detection Generalize?</i> , INTERSPEECH 2022
In-the-Wild (Müller et al.)	Baevski et al., <i>wav2vec 2.0</i> ; Babu et al., <i>XLS-R</i>
IndicVoices — genuine Indian multilingual speech	Tak et al., <i>RawBoost</i>
IndicSynth — 12-language Indian synthetic speech	Digital Personal Data Protection Act, 2023 — minimisation
MUSAN — noise corpus	
MLAAD — CC BY-NC, access requested	

USE THE LEFTOVER SPACE HERE — MOST TEAMS WASTE THIS SLIDE

Slide 6 is usually half empty. Put **Figure 9 (the roadmap)** in the remaining space. “Potential for future work progression” is one of the nine official criteria and almost nobody addresses it explicitly. It is also the *only* honest place to show the ambitious version of the system — gRPC, published SDKs, SIP connector, on-device quantised head, continuous retraining — clearly labelled as roadmap rather than as something built.

PART C — The diagram library

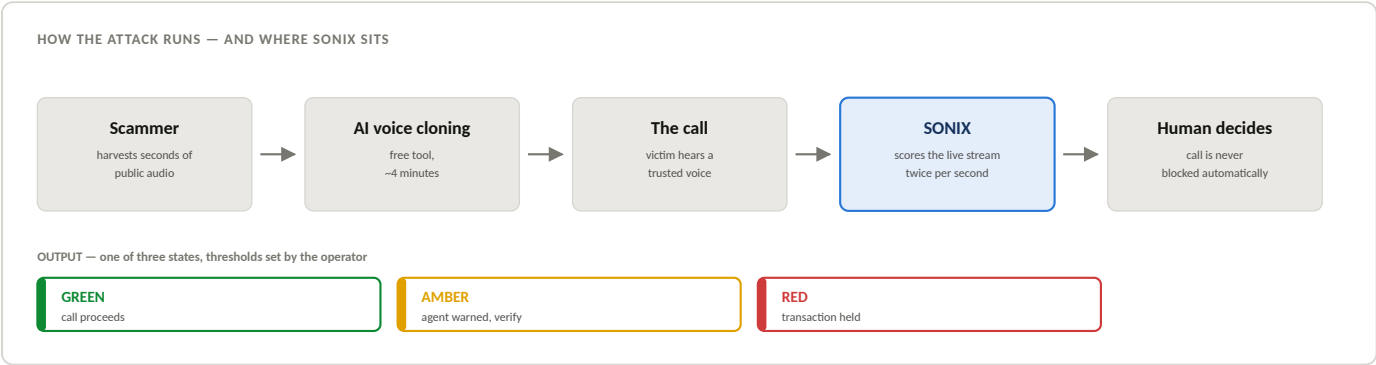
Eleven ready-to-use figures. Every one is supplied as a **standalone SVG** in `docs/deck_assets/` — vector, so it stays sharp at any size, and fully editable. PowerPoint: *Insert* → *Pictures* → *This Device*, pick the `.svg`, then right-click → *Convert to Shape* if you want to restyle individual elements. A 2×-resolution `.png` of each is included for anything that will not take SVG.

ONE RULE WHEN YOU RESTYLE THESE

Green, amber and red are reserved for the risk band. If you recolour the diagrams to a deck theme, do not introduce those three hues anywhere else — the moment a heading is green, the band stops reading as a status and starts reading as decoration.

SLIDE 2

Figure 1 — Attack and defence flow

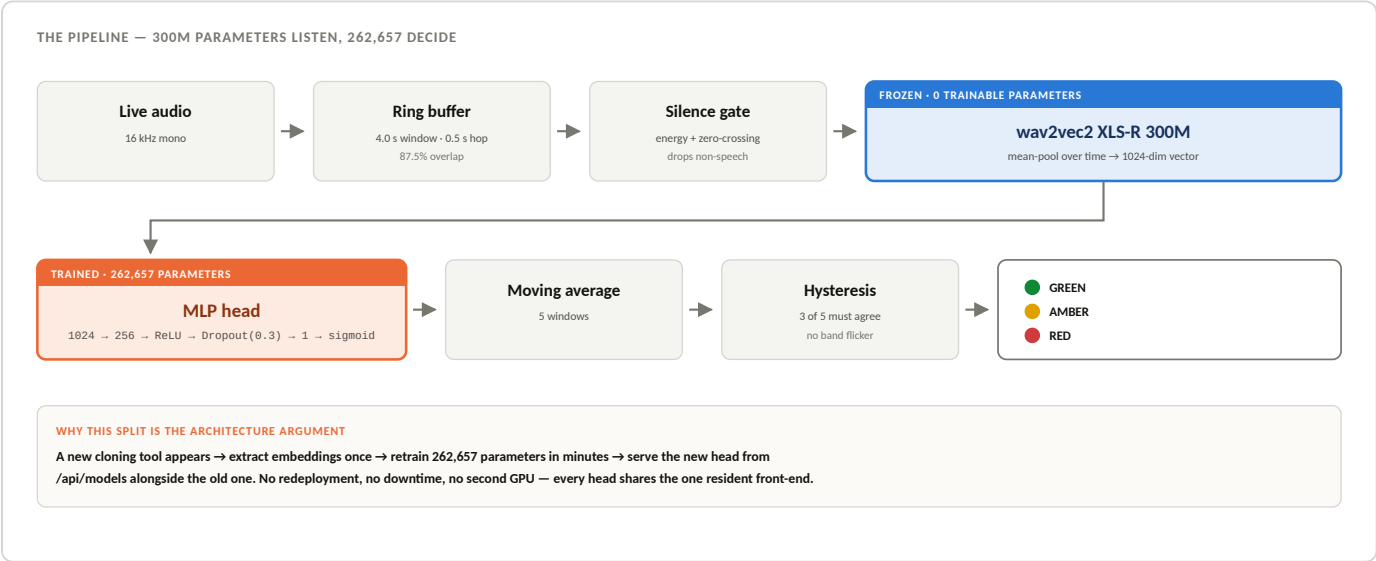


Caption to put under it in the deck: “SONIX never blocks a call. It gives a human a reason to pause.”

Why / how to use it: The five-step horizontal flow. Put the Green/Amber/Red strip on at least three slides — it is the product's visual identity and it makes the deck feel like one system.

SLIDE 3 (PRIMARY)

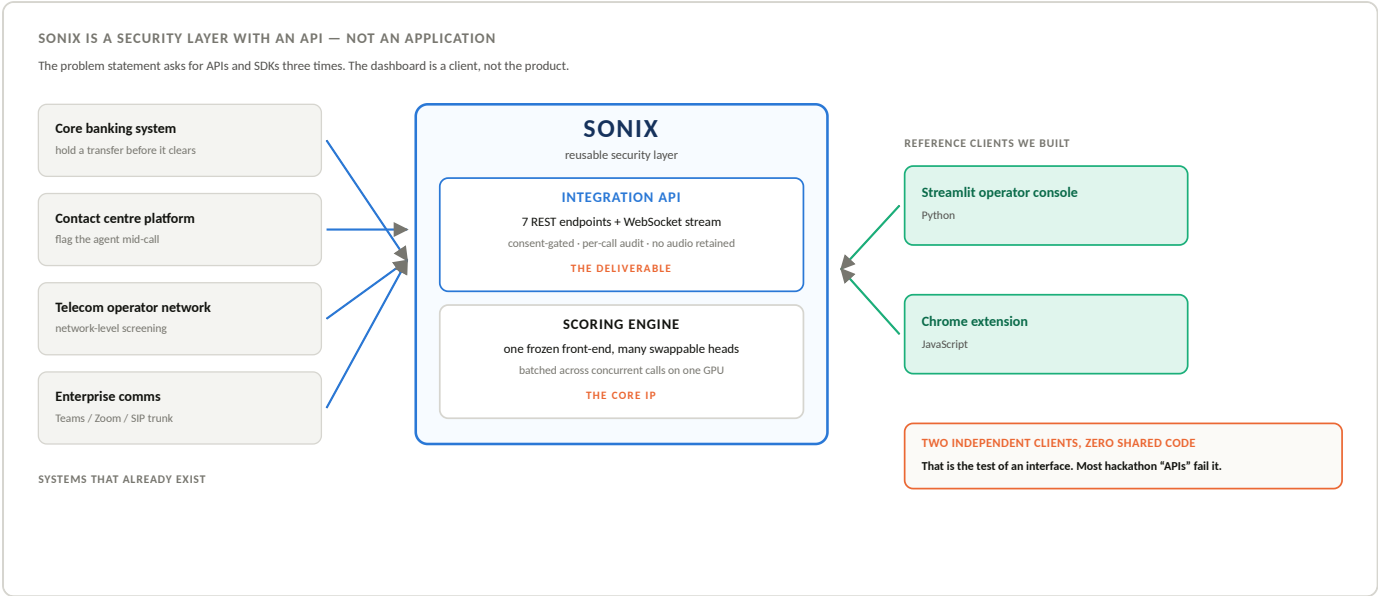
Figure 2 — The technical pipeline



Caption to put under it in the deck: “300 million frozen parameters listen. 262,657 trained parameters decide — so a new cloning tool costs minutes of retraining, not a release cycle.”

Why / how to use it: The hero diagram. The blue/orange FROZEN vs TRAINED split is the architecture argument made in one glance; do not recolour it to your theme, the contrast is doing the work.

Figure 3 — Integration architecture



Caption to put under it in the deck: “Two independent clients already consume the same endpoints — a Python dashboard and a JavaScript extension sharing no code. That is the test of an interface.”

Why / how to use it: This is the slide element that proves we read the problem statement. Pair it with the three-line SDK snippet.

Figure 8 — What exists today vs SONIX

WHAT EXISTS TODAY — AND WHAT SONIX DOES DIFFERENTLY			
	Published deepfake detectors	Commercial voice biometrics	SONIX
When does it run?	after the call, on a file	at enrolment / login	during the live call, updates 2x/second
What is shipped?	a research model	a closed product	REST + WebSocket API, 2 independent clients
New cloning tool appears	retrain the whole model	vendor release cycle	retrain 262k params in minutes, hot-swap
Indian languages	rarely evaluated	accent-sensitive	12 languages, both sides of the label
Consent + audit	not addressed	vendor-defined	enforced in the state machine, no audio kept
Failure behaviour	n/a	blocks / denies access	never blocks — flags for a human

Caption to put under it in the deck: “Every column is a design decision we can defend under questioning.”

Why / how to use it: This answers the template's 'innovation and uniqueness' field in a form a portal reviewer can scan in four seconds. A two-time SIH winner specifically recommends a competitor comparison when business people are on the panel.

SLIDE 4 (PRIMARY)

Figure 4 — Three shortcuts we found in our own model

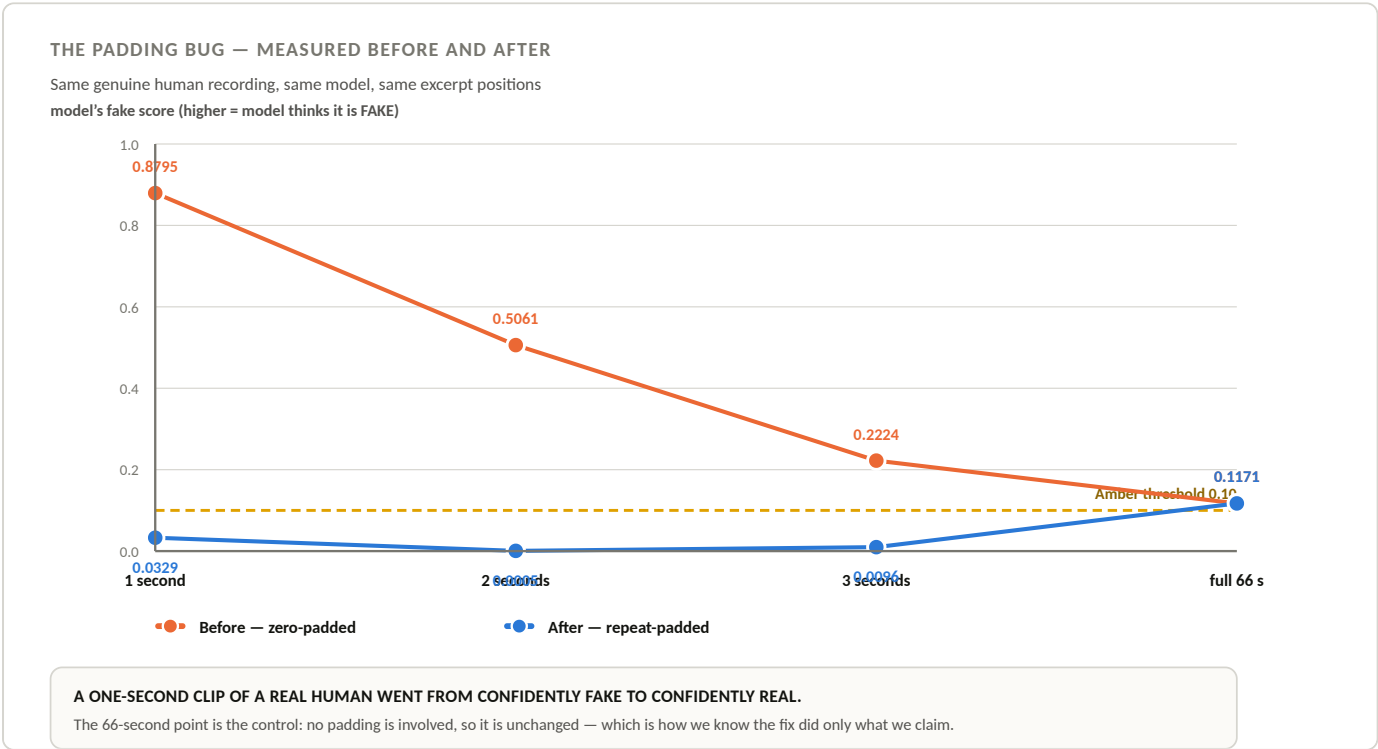
WE AUDIT OUR OWN MODEL — THREE SHORTCUTS WE FOUND IN IT				
Observation → hypothesis → measurement → root cause → fix in the DATA, not the threshold → verification				
OBSERVATION	WE MEASURED	ROOT CAUSE	FIX	RESULT
Real people flagged as fake	Score vs clip length	Short clips zero-padded with digital silence — a 1s clip was ¾ nothing	Repeat-pad the audio instead of padding with silence	0.8795 → 0.0329 FIXED
Genuine clips still flagged	Score vs signal-to-noise ratio	Model learned "clean background = fake". Training silence sits at ~75 dBFS — no real mic does that	Room reverb + real-world noise added to the training data	Failures 39–67 dB passes 35–38 dB FIXED
Genuine Indian speech flagged	Matched real/clone pair, same speaker	Our Indian corpus is bonafide-ONLY — so it learned "Indian audio = real", not detection	Built 35,200 Indian synthetic clips across 3 generator families, 12 languages	Verification in progress IN VERIFICATION

Caption to put under it in the deck: “Found by us, measured by us, fixed in the training data — not by tuning a threshold. The third is still being verified, and we say so.”

Why / how to use it: The rarest thing in the deck. Almost no team shows a measured before-and-after of a bug they found themselves. Keep the amber 'IN VERIFICATION' badge exactly as it is — the honesty is the point, and the official template requires a risks field anyway.

SLIDE 4 (INSET) OR BACKUP

Figure 6 — The padding bug, measured

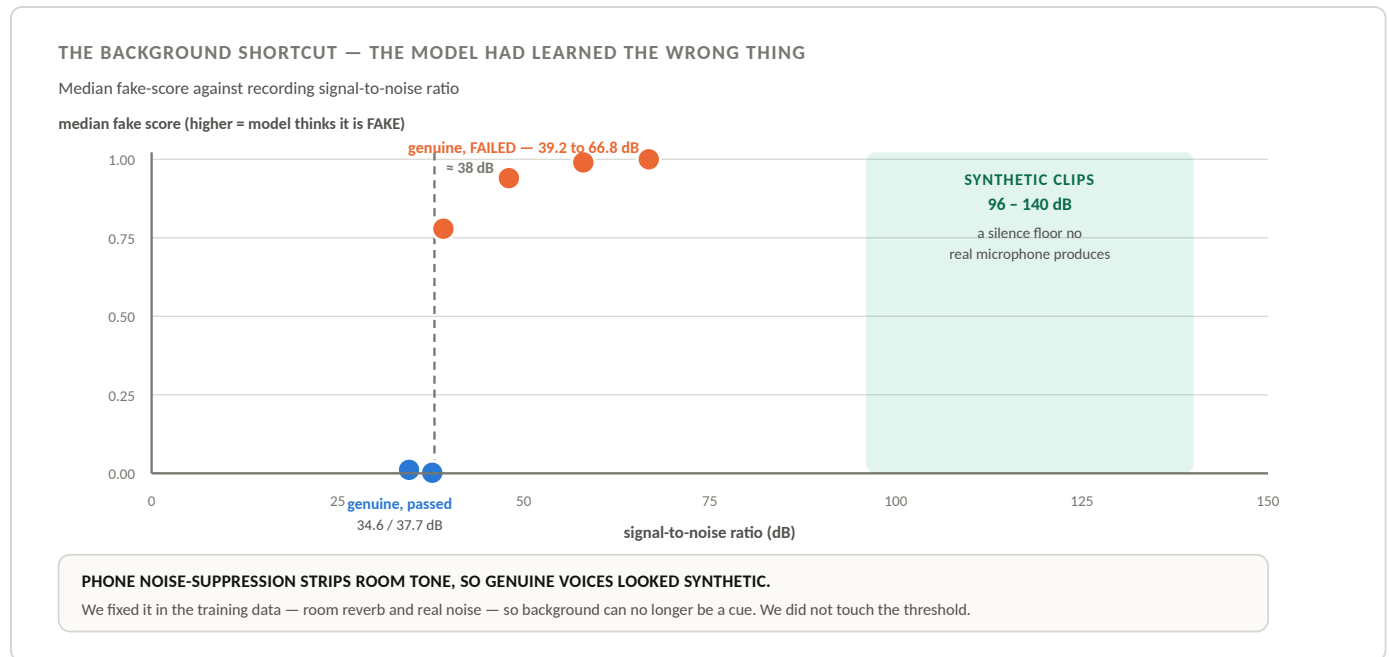


Caption to put under it in the deck: “A one-second clip of a real human went from confidently fake to confidently real.”

Why / how to use it: The 66-second control point is what a sharp judge will respect: no padding is involved there, so it is unchanged, which proves the fix did only what we claim.

SLIDE 4 (INSET) OR BACKUP

Figure 7 — The background shortcut, measured



Caption to put under it in the deck: “The model had learned ‘clean background = fake’, not synthesis detection. We found it, proved it, and fixed it in the training data.”

Why / how to use it: If slide 4 is crowded, Figure 4 already summarises both charts — hold 6 and 7 as backup slides for the oral rounds, where a judge who asks ‘how did you find that?’ gets the evidence.

SLIDE 2 OR SLIDE 4

Figure 5 — Indian languages on both sides of the label

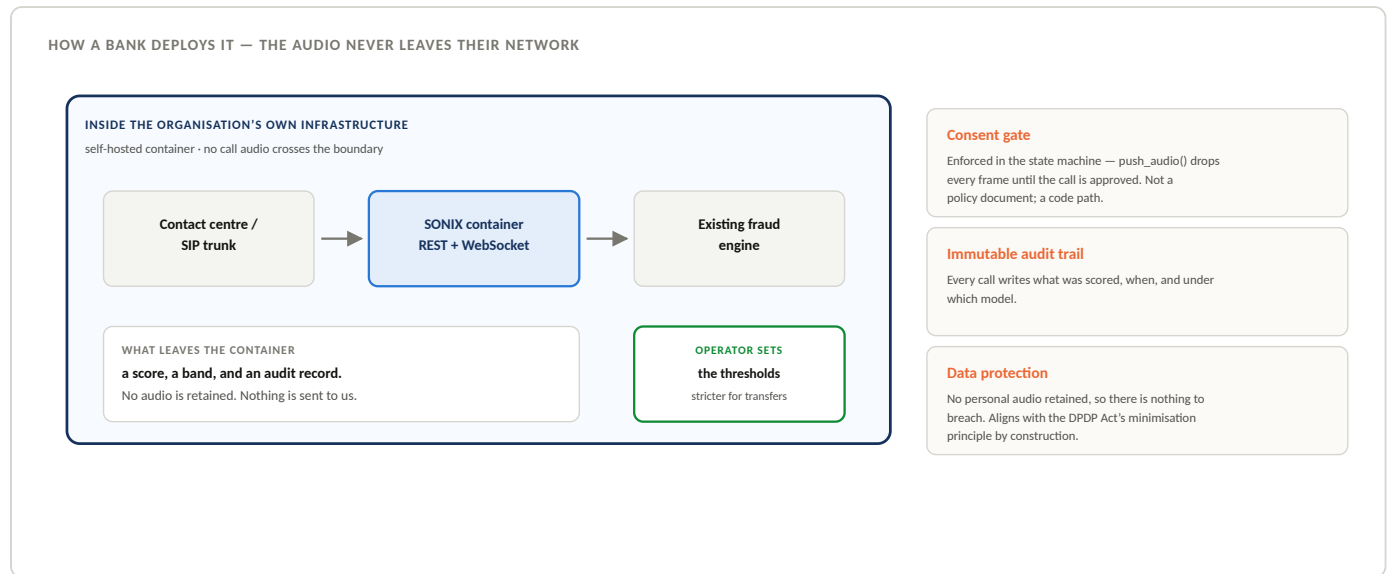


Caption to put under it in the deck: “37,032 genuine to zero synthetic was the bug. Now it is 1.05 : 1.”

Why / how to use it: The single clearest picture of the dataset contribution. If a judge only looks at one chart, this is the one that shows we built something rather than downloaded something.

SLIDE 5 (CHOOSE THIS OR FIG 11)

Figure 10 — Deployment and privacy

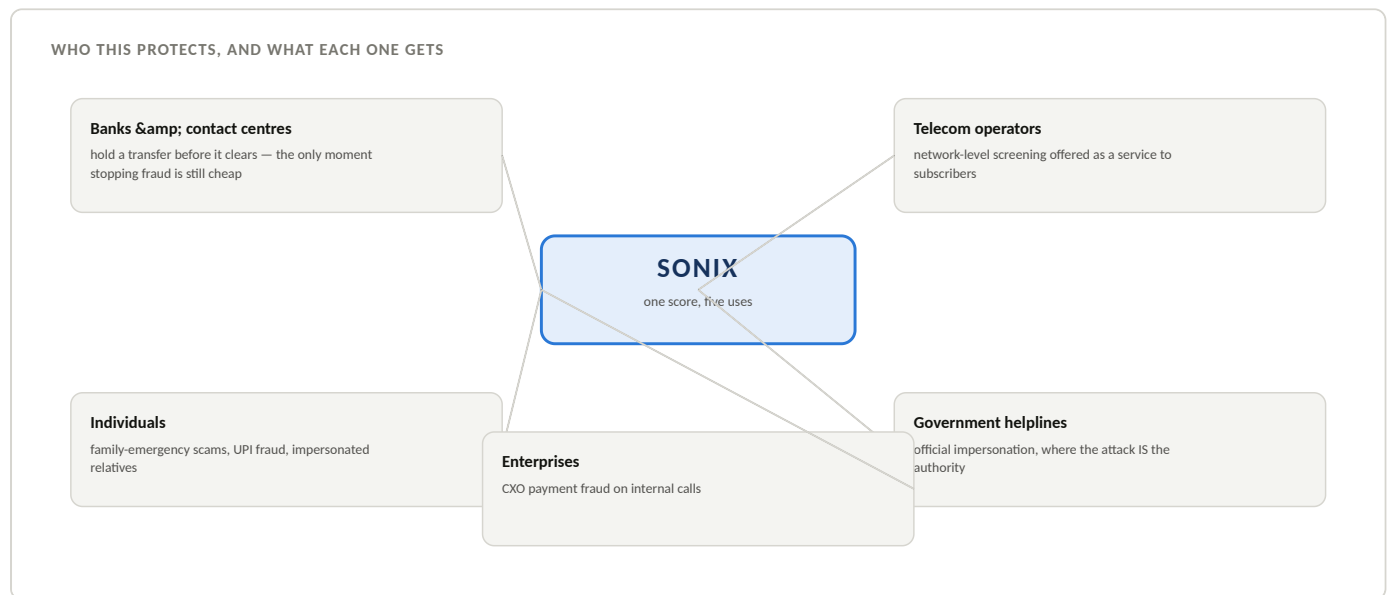


Caption to put under it in the deck: “Self-hosted. The call audio never leaves the organisation's network — only a score and an audit record.”

Why / how to use it: Use this one if the jury skews technical or government. 'Practicability' and 'sustainability' are separate official criteria, and this diagram answers both.

SLIDE 5 (ALTERNATIVE)

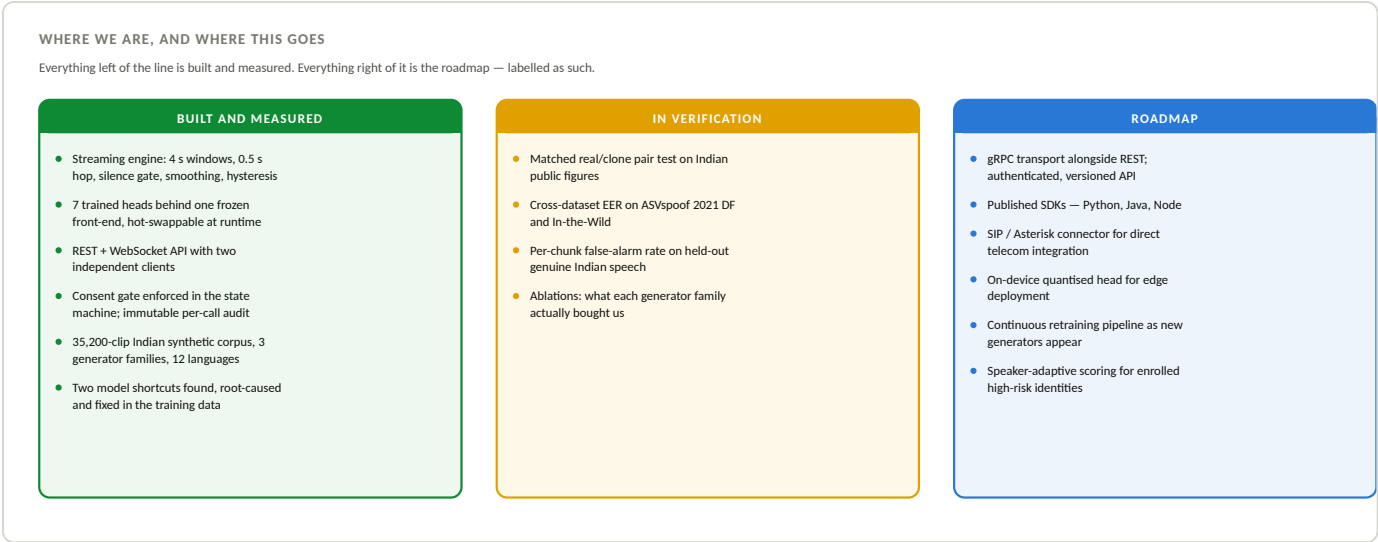
Figure 11 — Who this protects



Caption to put under it in the deck: “One score, five different uses — each with a named owner who already exists inside the organisation.”

Why / how to use it: Use this if the jury skews business or social. Do not use both 10 and 11 — one primary visual per slide.

Figure 9 — Where we are and where this goes



Caption to put under it in the deck: “Everything in green is built and measured. Everything in blue is roadmap, and it is labelled as roadmap.”

Why / how to use it: 'Potential for future work progression' is one of the nine official criteria and almost nobody addresses it. This is also the ONLY honest place to show the ambitious version of the system — the three bands keep the distinction visible, which is what protects every claim to its left.

PART D — What actually wins at the national round

Researched 13 September 2026 from official SIH sources and first-hand winner accounts. Every claim below is marked **verified** (official SIH document or primary winner account) or **single-source** (one credible practitioner, uncorroborated). Sources are listed at the end of this part.

FIRST: MOST “SIH GUIDE” CONTENT ON THE OPEN WEB IS FABRICATED

The top search result for “SIH PPT template” states the deck is **ten slides** and publishes a percentage rubric. Both are false — the official template says six, and SIH publishes no weights at all. A second popular blog publishes a *different* invented rubric. Two confident, mutually contradictory rubrics, neither sourced. **We were using one of them.** Discard both.

D1. The official evaluation criteria — verified

Verbatim from the SIH FAQs and repeated in the official SPOC guidelines. Nine named criteria, **no numeric weights published anywhere**:

“novelty of the idea, complexity, clarity and details in the prescribed format, feasibility, practicability, sustainability, the scale of impact, user experience and potential for future work progression.”

Three of those deserve attention because they are unusual and most teams ignore them:

- **“clarity and details in the prescribed format”** — template compliance is not hygiene, it is a scored line. Wrong slide count, altered pointers, or leftover placeholder text costs marks directly.
- **“practicability” and “sustainability”** — listed *separately* from “feasibility”. That is asking: can this actually be run, by whom, and does it keep working? Self-hosted deployment, consumer-GPU training and minutes-not-days retraining all answer this.
- **“potential for future work progression”** — roadmap is an official criterion. This is where the ambitious version of the system legitimately belongs.

D2. How the finale actually runs — verified, and the most actionable finding

Judging is **multi-round with the same jury**, not a single pitch. From the official Nodal Center Organizer Manual: *“The panel of 3/5 days judges who have already given feedback to teams will be in jury panel”*, with a feedback session once per day plus a **power judging session on the final day**.

A 2025 nodal-centre report documents three rounds with distinct focus:

ROUND	WHAT IT SCORES
1 — Day 1 evening	solution presentation, innovation, feasibility, execution planning
2 — Day 2 morning	prototype development, improvements, integration, usability, teamwork
3 — Day 2 evening (power round)	functionality, final demonstration, user experience, market readiness, implementation scope

THE IMPLICATION — THIS CHANGES HOW WE PREPARE

Judges score the delta between visits. The same three people see you three times. A team that arrives with a finished thing and shows the same finished thing three times scores worse than a team that visibly acts on feedback.

Deliberately hold something back — land the `sonix_real/` pair-test result, or a threshold change a judge suggested, between round 1 and round 2. A 2019 winner: “We were open to change the whole time and constantly interacted with the judges to get to know what they want.” The 2024 first-place team's account is the same: three rounds, and they “took the feedback seriously and made key changes.”

D3. What winners say differentiated them — verified quotes

BUSINESS VALUE OVER TECHNOLOGY — THE MOST REPEATED THEME

A finale winner on a corporate PS was told by judges to “think more about how this will benefit the corporate and **focus less on the tech aspect of it,**” and concluded: “**The judges were less concerned about the technology and more about the solution.**” They were also warned that the power round “matters a lot and is more or less **make or break.**”

For us: lead with the transfer being held, not with wav2vec2. The architecture is the *reason* we can do it, not the pitch.

JUSTIFICATION DENSITY

A two-time winner (2022, 2023): “**Every small thing on your project should be backed by reason.**” She also puts the novelty bar at “at least 30 % uniqueness” versus existing solutions, and recommends a competitor analysis when business people are on the panel — which is exactly what Figure 8 is for.

BE READY TO PROVE ORIGINALITY

A 2023 winner: “Round 1: We impressed the judges, but then they **suddenly accused us of copying the frontend.**” They survived by producing original Figma files. **Bring provenance** — our git history with per-commit messages, the corpus generation scripts, the training logs. For us the sharper version of this question is “*this is a wrapper on somebody else's model, what is actually yours?*” and the answer is in §E.

D4. Why teams lose — documented failure modes

FAILURE	SOURCE STRENGTH	OUR EXPOSURE
Plagiarism suspicion on UI or assets	verified	medium — bring git history and generation scripts
Integration breaks before the final round	verified	high — freeze code 2 h before; backup video
Not fully solving the stated problem	verified	low — but re-read the PS aloud before each round
One member cannot justify a technical decision under questioning	verified	high — see §E, everyone owns their section
Unprepared for follow-ups about flaws or pivots	verified	low — this is our strongest ground
Template placeholder text left on a slide; theme metadata mismatched	single-source	trivially avoidable — check before export
Demo on a free tier that sleeps, blank page when the judge opens it	single-source	n/a — ours is local, which is an advantage: say so
Saying “dummy data” instead of naming a real source	single-source	low — every dataset we use is named

D5. Honest framing versus polished claims — the evidence

No official source addresses this directly, but three independent sources point the same way and one is a winner.

- A practitioner guide to the finale: **“Nobody believes a 36-hour build is bulletproof, so claiming it is costs you more than the flaws would.”** It lists as live judge questions: “What is your accuracy, and how did you measure it?”, “How much of this did AI write?”, and “This is a wrapper on somebody else’s API. What is actually yours?” — and flags **overconfidence** as the critical trap: judges test whether teams recognise their own weaknesses. (single-source)
- A two-time winner: teams should **“demonstrate transparency about journey, hurdles, and pivots”**, and lists as a losing mistake **“lack of preparedness for follow-up questions about flaws or pivots.”** (verified)
- Structural proof: the official template **mandates** a “Potential challenges and risks” field. The rubric requires you to state what is hard. (verified)
- An actual **SIH 2024 winning deck on a security problem statement** wrote **“80 % completed with testing/validation pending”** on its own technical slide — and won. (verified artifact)

THE ONE REAL TENSION, AND HOW TO RESOLVE IT

The same two-time winner also advises making the product **“visually complete rather than work-in-progress”**. These reconcile cleanly: **the artifact should look finished; the claims about it should be measured.** A polished dashboard with an honest benchmark table beats both a scrappy demo and an inflated number. Honest framing belongs in the *spoken defence and the stated metrics* — not in a deck that looks half-built.

D6. What the 2024 winning security deck actually did — worth copying

- **Named and coined its mechanism** rather than its product: “Time-based One-time Handshake.” The innovation was a named primitive. → For us: name the architecture. **“Frozen listener, swappable judge.”** A judge who repeats your phrase back to you in round 3 has adopted your framing.
- **An explicit differentiator list against the incumbent** — enumerated, not prose. → Figure 8.
- **A degradation path as a headline feature** — offline auth, IMEI fallback for basic phones, works with no internet. Indian-infrastructure realism on slide 2. → For us: self-hosted, works on a 2 Mbps link because we ship scores not audio, runs on consumer GPUs.
- **A stated completion percentage** rather than a claim of done.
- **Peer-reviewed references** on slide 6.
- Notably **absent**: no cost table, no business-model canvas, no patent slide. With six slides those do not fit — hold them for the oral rounds.

D7. Compliance and Indian-infrastructure framing — weak evidence, stated honestly

We could find **no** first-hand account from a team on a cybersecurity PS describing jury questions, and **no** official SIH document mentioning CERT-In, DPDP, TRAI or RBI. Treat this section as a hypothesis.

The strongest data point is a practitioner guide listing as a standard finale question: “What law or policy does this have to comply with?” with the recommended answer — *name one that genuinely applies and what you did about it; the DPDP Act if you touch personal data*. Related documented questions: “Who inside the ministry owns this on day one?”, “Which of our existing systems would this have to talk to?”, “What is the smallest version we could deploy next month?”

Circumstantial support that deployment-realism is rewarded **is** solid: the 2025 finale jury praised solutions with **“remarkable potential for real-world deployment”**; round 3 explicitly scores “market readiness and implementation scope”; and “practicability” and “sustainability” are separate official criteria.

Our position: name DPDP once, on slide 5, tied to a concrete design decision (no audio retained). Prepare the three deployment-ownership answers in §E. Do not carpet-bomb the deck with acronyms.

D8. Scale context

SIH 2025's software edition ran across **60 nodal centres with roughly 8,000 students**, five winning teams per centre-category. The shortlist quota per college has grown — a 2025 notice shows “45 shortlisted + 5 waitlisted”. **We could find no source describing what changes when a waitlisted team is called up**, or whether the national submission expectation differs. Assume it does not, and submit as though you were shortlisted outright.

SOURCES

Official: sih.gov.in FAQs · Nodal Center Organizer Manual and Guidelines · SIH2024 Idea Presentation Format (CMRIT mirror) · SIH2025 template · SPOC guidelines (NIT Jamshedpur mirror) · SIH 2025 finale nodal-centre report (Swaminarayan University).

Winner accounts: Chethan AC (2024, 1st place) · Sarika Purohit (2023) · Ishika Jain (2022 & 2023) · Tanmay Bhatnagar (finale) · SIH 2024 winning deck “CODESTRIX” (SlideShare).

Single-source practitioner guides: zaidsayyed.in/blog/sih-2026 · velonx.in SIH 2026 guide.

Disconfirmed: reskilll.com 10-slide/percentage-rubric claim · anishprashun.me rubric.

PART E — The Q&A bank

Three judges, three rounds. **Assign an owner to every answer** — one member answering everything is a documented failure mode.

“WHAT IS YOUR ACCURACY, AND HOW DID YOU MEASURE IT?”

Yugal. “We report EER, not accuracy — the benchmark data is about 90 % spoof, so a model that says ‘fake’ every time scores 90 % accuracy and is useless. In-domain we get `[[EER_INDOMAIN]]` on the ASVspoof 2019 eval set, 71,237 clips, and that number was reproduced independently by two implementations of the metric. Cross-dataset it degrades, and I can show you the exact figure and the set it came from.”

“THIS IS A WRAPPER ON SOMEBODY ELSE'S MODEL. WHAT IS ACTUALLY YOURS?”

Yugal. “The front-end is public and frozen — we use it as a feature extractor and we say so. What is ours is four things. The trained head and its training recipe. The streaming architecture — windowing, silence gate, smoothing, hysteresis — none of which exists in the published detector we started from. The API and the two clients on it. And a 35,200-clip Indian synthetic speech corpus across three generator families that we generated, augmented and labelled ourselves, because it did not exist. That last one is a dataset contribution, not a wrapper.”

“CAN IT DETECT A CLONED INDIAN-LANGUAGE VOICE?”

Yugal. “That is our next measurement and I will not claim it yet. Here is what we know: we proved we do not false-alarm on genuine Indian speech, and we catch Western synthetic speech. We then found that our Indian-language model had learned the wrong thing — every Indian clip it had seen was genuine, so it learned ‘Indian audio is real’. We built the missing half of that data. The verification is running.”

⚠ Do not claim it works until the `sonix_real/` test passes. If it has passed by the round, this becomes the strongest answer in the deck.

“HOW WOULD A BANK ACTUALLY DEPLOY THIS?”

Akshat. “Self-hosted container inside their own infrastructure — the call audio never leaves their network. They call our REST endpoint or open a WebSocket from their contact-centre platform. Thresholds are configured per workflow: stricter on a fund transfer than on a general enquiry. The console we are showing you is a reference client; in production the score feeds their existing fraud engine.”

“WHO INSIDE THE ORGANISATION OWNS THIS ON DAY ONE?”

Akshat. “For a bank, the fraud-risk operations team — they already run rules on transactions, this is one more signal into the same queue. For a telecom operator, the network security operations centre. The person who acts on a red band already exists; we are not asking anyone to create a new role.”

“WHAT IS THE SMALLEST VERSION YOU COULD DEPLOY NEXT MONTH?”

Yugal. “One container beside an existing contact-centre recorder, scoring a sampled fraction of calls, writing the band into their case-management system. No integration into the call path at all, so nothing can break a live call. That gets real-world data flowing and it is a two-week engagement, not a programme.”

“WHAT LAW OR POLICY DOES THIS HAVE TO COMPLY WITH?”

Navya. “The DPDP Act, because call audio is personal data. Our design answer is minimisation by construction — we retain no audio at all, only a score and an audit record, and consent is enforced in the state machine rather than in a policy document. There is a code path that drops audio frames until the call is approved.”

“WHY EER AND NOT ACCURACY?” / “WHY NOT JUST USE A THRESHOLD OF 0.5?”

Yukti. “EER balances false accepts against false rejects, and with ~90 % spoof data accuracy is meaningless. On the threshold: 0.5 is arbitrary. Ours came from the eval score distribution — amber at 0.10, red at 0.90 — and the operator can move both, because the cost of a false alarm on a fund transfer is different from a general enquiry.”

“WHAT HAPPENS WHEN A NEW CLONING TOOL APPEARS?”

Yugal. “The front-end is frozen and the head is 262,657 parameters. We extract embeddings once and retrain in minutes, then serve the new head from `/api/models` alongside the old one so they can be compared. No redeployment, no downtime. That was an architecture choice made specifically because the threat landscape changes weekly.”

“WHAT IS YOUR COST PER CALL?”

Yugal. **PREPARE THIS — \$F ITEM 5** Being roughly right with your reasoning visible beats a confident number you cannot break down. Work out GPU-seconds per call-minute from the batch throughput and price it off a public cloud GPU rate. Have the arithmetic on a card.

“HOW MUCH OF THIS DID AI WRITE?”

Yugal. Answer plainly — you used AI assistance, as most teams did, and you can explain any line in the repo. Then demonstrate it: open `engine.py` and walk the batching logic, or explain why `_trim_pending` broke. The defence against this question is not denial, it is fluency.

“WHAT IS YOUR BIGGEST WEAKNESS?”

Yugal. “Cross-dataset generalisation, and we measure it rather than hide it. Second is that our Indian-clone detection is built but not yet verified — we will have that number before the final round.”

THE THREE SENTENCES THAT SAVE A ROUND

Wrong number in your head: “I would rather check that than guess — it is in our results table.”

Demo fails: “That is the live path failing. Here is the recorded run, and here is what is happening underneath.”

Question you cannot answer: “We have not measured that. Here is how we would.”

To a practitioner jury all three read as competence, not weakness.

PART F — What must be run before submission

In priority order. The first two are cheap and they unlock slide 4.

#	TASK	OWNER	WHY IT MATTERS
1	The sonix_real/ matched-pair test. Run every head over the real/clone pairs. <code>python bench_clips.py --dir sonix_real\sonix_real --all-models</code>	Yugal	Decides whether the entire Indic effort worked, which model ships, and whether the Indian-clone claim can be made at all. Fills <code>[[INDIC_CLONE_DETECT]]</code> . This is the gate.
2	verify_head.py on the current checkpoints	Yugal	The 1.4937 % headline needs to survive the merge overwrite before it goes back on a slide. Fills <code>[[EER_INDOMAIN]]</code> .
3	Score the IndicVoices holdout with <code>v3</code> , then <code>flag_rate.py</code>	Navya	Fills <code>[[FA_INDIC]]</code> . Must be reported per 4-second chunk , not per call.
4	Cross-dataset EER on DF21 for the final model	Navya	Fills <code>[[EER_DF21]]</code> . Must carry “(75 % key coverage)”.
5	Cost and scale. GPU-seconds per call-minute, concurrent calls per instance, monthly cost at 10k calls/day	unassigned	30 minutes of arithmetic. Winners consistently cite financial literacy as the differentiator, and it is a documented jury question.
6	Ablations: no-Indic-spoof / MMS-TTS only / IndicSynth only / no augmented copies	Yugal	Turns “we built a corpus” into “we measured what the corpus bought”. Strong round-2 material.
7	Record the backup demo video from a clean run	Suryansh	Integration breaking before the final round is a documented failure mode. If the laptop dies, the video is the demo.
8	Export the deck to PDF , check every template field against the portal, delete the “Important Pointers” slide	Yukti	“Clarity and details in the prescribed format” is a scored criterion. No PPT or DOCX will be accepted.

IF YOU ONLY DO TWO THINGS

Run **item 1** and **item 2**. Between them they convert three `[[PLACEHOLDER]]` tokens into real numbers on the strongest slide in the deck, and they cost an afternoon. Everything else on this list is an improvement; those two are the difference between a benchmark table with results and one with promises.

Final checklist

- Official SIH 2026 template downloaded from the portal and used unmodified — **six slides**, exported to **PDF**
- “Important Pointers” slide deleted; no placeholder text anywhere; theme and category match the portal
- Every slide readable **with the sound off** — the portal round has no presenter
- One primary visual per slide; diagrams over paragraphs; max six bullets
- “Higher score = model thinks it is fake” stated wherever a score appears
- Every DF21 figure carries “(75 % key coverage)”
- No blank cells — the benchmark matrix shows status markers instead
- Every `[[PLACEHOLDER]]` either filled by its owner or the row cut
- Green/Amber/Red used **only** for the risk band, nowhere else in the deck
- Roadmap on slide 6, clearly labelled as roadmap

- Backup video recorded; code frozen two hours before; HF_HUB_OFFLINE=1 set
- Every Q&A answer has a named owner and that person has said it out loud